

Generation Interconnection System Impact Study Report

For Queue Project Q22-1984

Taylorville South 230 kV

December 18, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1984. The Interconnection Customer proposes a 1247.0 MW natural gas combined cycle facility interconnecting at the Taylorville South 230 kV substation in the Illinois area. The request was received on January 15, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 700 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$138,789,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: INTERBAY – SHADY OKS WIND FARM 230.0 kV Line
Contingency: PHELPS DODGE – EASY ROAD 230.0 kV line
Loading: 183.4% (Rating: 550 MVA)
DFAX: 20.9%
Pre-project loading: 136.1%
2. Facility: PHELPS DODGE – EASY ROAD 230.0 kV
Contingency: SHADY OAKS WIND FARM – INTERBAY 230.0 kV L
Loading: 157.1% (Rating: 733 MVA)
DFAX: 35.6%
Pre-project loading: 96.5%
3. Facility: PHELPS DODGE – NRG ROCKORD I SUBSTATION 230.0 kV
Contingency: EASY ROAD – PHELPS DODGE 230.0 kV line
Loading: 132.9% (Rating: 711 MVA)
DFAX: 30.0%
Pre-project loading: 80.3%
4. Facility: NRG ROCKFORD I – LAS GALLINAS 230.0 kV
Contingency: NRG ROCKFORD I – MAZOMANIE 230.0 kV Line
Loading: 117.6% (Rating: 631 MVA)
DFAX: 17.5%
Pre-project loading: 83.0%
5. Facility: ALDER – SHADY OAKS WIND FARM 230.0 kV
Contingency: SHADY OAKS WIND FARM SUBSTATION – ALDER 500.0 kV line
Loading: 116.3% (Rating: 1,377 MVA)
DFAX: 14.1%
Pre-project loading: 103.6%
6. Facility: IVYGLEN METER – ALDER 230.0 kV Cir. 1
Contingency: FORT ST VRAIN – ALDER 500.0 kV
Loading: 101.5% (Rating: 718 MVA)

DFAX: 9.6%
Pre-project loading: 84.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
INTERBAY – SHADY OKS WIND FARM 230.0 kV Line	Rebuild line with higher-capacity conductor	\$19,334,000	50% (\$9,653,000)
PHELPS DODGE – EASY RD 230.0 kV	Rebuild line with higher-capacity conductor	\$31,829,000	62% (\$19,616,000)
PHELPS DODGE – NRG ROCKORD I SUBSTATION 230.0 kV	Rebuild line with higher-capacity conductor	\$139,807,000	66% (\$92,269,000)
NRG ROCKFORD I – LAS GALLINAS 230.0 kV	Rebuild line with higher-capacity conductor	\$7,676,000	54% (\$4,122,000)
ALDER – SHADY OAKS WIND FARM 230.0 kV	Rebuild line with higher-capacity conductor	\$19,313,000	32% (\$6,224,000)
IVYGLEN METER – ALDER 230.0 kV #1	Reconductor line	\$19,565,000	35% (\$6,905,000)
	Total allocated to project		\$138,789,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.